
Differentially Private Preference Data Synthesis for Large Language Model Alignment

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Abstract

Preference alignment constitutes an indispensable post-training mechanism for large language models (LLMs), insofar as it calibrates model behavior toward human-valued responses. Yet post-training on authentic human preference corpora introduces acute privacy vulnerabilities, because such corpora commonly encode sensitive user prompts together with annotators’ evaluative judgments. To address this tension, we propose **DPPrefSyn**, a new algorithmic framework for constructing differentially private (DP) synthetic preference data that enables privacy-preserving preference alignment. DPPrefSyn is grounded simultaneously in the Bradley–Terry preference model and in the intrinsic geometric organization of pairwise human preference observations. The method first estimates an underlying preference model from private data under formal differential privacy guarantees, after which it combines this protected model with public prompts to synthesize high-quality preference triples. By exploiting the shared linear structure of per-cluster reward models, DPPrefSyn captures heterogeneous human preferences in private datasets, while DP Principal Component Analysis (DP-PCA) improves the statistical efficiency of subsequent learning. Extensive experiments demonstrate that DPPrefSyn attains competitive alignment utility even under stringent DP constraints, thereby indicating that synthetic preference data can serve as a practical substitute for direct use of private preference corpora across a broad spectrum of alignment settings. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

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1. Introduction

Preference alignment techniques, including RLHF (Stienon et al., 2020) and DPO (Rafailov et al., 2024), have become central instruments for bringing LLM behavior into closer correspondence with human expectations. These algorithms typically depend on pairwise preference data, in which human annotators compare two candidate responses and identify the one that more adequately satisfies the prompt. The resulting comparisons are then used to fine-tune LLMs by inducing a ranking preference for selected responses over rejected alternatives (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). Because preference alignment has proved effective and reliable in practice, it has been widely adopted in systems such as chat assistants (Achiam et al., 2023), mathematical reasoning tools (Shao et al., 2024), and code generators (Shen et al., 2023).

Nevertheless, the use of human preference data for LLM alignment creates substantial privacy concerns, since these datasets frequently contain real user prompts and direct human feedback. The prompts themselves may disclose personal information concerning health, identity, or other sensitive domains, while the corresponding judgments may reveal private beliefs, preferences, or behavioral regularities (Li et al., 2023; Yu et al., 2024). Prior work (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a) has investigated the mitigation of these risks through the rigorous protection afforded by differential privacy (DP) (Dwork et al., 2006b), yielding encouraging initial evidence. However, many existing approaches protect only a subset of the relevant data channels—for instance, privatizing user prompts (Yu et al., 2024) or preference labels (Chowdhury et al., 2024; Zhang et al., 2025), but not both simultaneously. Such partial protection can leave residual privacy exposure, because adversaries may still infer sensitive attributes from unprotected prompts or annotations. In addition, existing methods are often tied to particular post-training procedures, such as RLHF (Wu et al., 2023a), and therefore do not naturally accommodate newer alignment methods such as DPO. Finally, because private preference datasets are costly and difficult to scale, these methods may be constrained by limited annotation resources and consequently fail to deliver high-quality preference alignment. These limitations

motivate the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?

In this work, we answer this question affirmatively. Our main contributions are as follows:

- We introduce DPPrefSyn (Figure 1), a new algorithm for generating DP synthetic preference data for LLM alignment. DPPrefSyn provides a principled framework that couples the Bradley–Terry preference model with the intrinsic geometry of pairwise human preference observations, and it incorporates three central design elements: 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering*. The Bradley–Terry preference model, together with a linear reward structure, permits preference samples to be grouped according to embedding differences between concatenated prompt–preferred and prompt–dispreferred pairs, so that preferences within each cluster can be approximated by a shared reward function. 2) *Improving learning efficiency through DP-PCA*. To mitigate sample inefficiency and clustering instability in high-dimensional representation spaces, we apply DP-PCA, which projects embeddings into a lower-dimensional subspace while preserving salient preference signals under DP. 3) *Saving privacy budgets by using public prompts*. By generating candidate responses from public prompts and using private reward models only for preference-pair selection, we concentrate the privacy budget on the modeling of private preferences and thereby improve privacy efficiency.
- We provide a rigorous privacy analysis establishing that DPPrefSyn satisfies (ϵ, δ) -DP. The analysis allocates a privacy budget ϵ_0 to DP-PCA (Amin et al., 2019) for dimensionality reduction and ϵ_1 to DP-KMeans (Su et al., 2016) for clustering, while reserving the remaining budget for the training of per-cluster reward models via DP-SGD. We then use the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to tightly compose the privacy cost of DP-SGD, thereby ensuring that DPPrefSyn satisfies (ϵ, δ) -DP. Owing to the post-processing property of DP, the resulting synthetic dataset can subsequently be reused across multiple preference-alignment methods and different LLMs without additional privacy expenditure.
- We empirically evaluate DPPrefSyn on standard benchmarks, including question-answering tasks from OpenAssistant (Köpf et al., 2023) and Anthropic-HH (Bai et al., 2022), as well as the TL;DR summarization task (Stiennon et al., 2020). The experiments indicate that DPPrefSyn-generated synthetic preference data delivers strong privacy protection while preserving competitive utility across heterogeneous tasks. For example, with $\epsilon = 2$, DPO

fine-tuning of Pythia-2.8B on our synthetic data attains a GPT-4o win rate of 56.48% on Anthropic-HH, substantially exceeding the 37.02% obtained by fine-tuning on real data. These results demonstrate that DP synthetic preference data can support effective LLM preference alignment under strong privacy guarantees. Moreover, because DPPrefSyn can be used with multiple preference optimization methods (e.g., DPO and RLHF) and remains effective under stringent privacy constraints, it avoids the method-specific limitations of several existing privacy-preserving post-training approaches.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) introduce a DP framework for aligning LLMs with reinforcement learning by adapting PPO to the DP setting; however, their approach remains *limited to PPO*, even as preference alignment continues to diversify methodologically. Chowdhury et al. (2024) study reward estimation from preference-based feedback under label-DP (Ghazi et al., 2021), thereby protecting human annotators’ labels, and Zhang et al. (2025) propose AUP-RLHF, a user-level label-DP framework for RLHF. Although these methods can safeguard preference labels, they *do not address the sensitivity of prompts or responses themselves*. Moreover, because such methods are usually *constrained by the limited private user preference data*, they may not achieve high-quality preference alignment when human preference annotations are expensive to obtain at scale. Yu et al. (2024) instead focus on protecting sensitive user *instructions* by generating DP synthetic *instructions* to replace real instructions during data annotation and model fine-tuning.

Beyond preference alignment, another line of work investigates private LLM fine-tuning with general labeled datasets rather than preference comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022), while recent studies on in-context learning examine mechanisms for protecting the privacy of prompts supplied at inference time (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b).

Differentially Private Synthetic Text Generation. Our work is also situated within the broader literature on DP synthetic text generation, where two representative paradigms have emerged. The first privately fine-tunes a language model on sensitive data and subsequently uses the fine-tuned model to produce synthetic text (Yue et al., 2023; Mattern et al., 2022; Mireshghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and then use the resulting model to

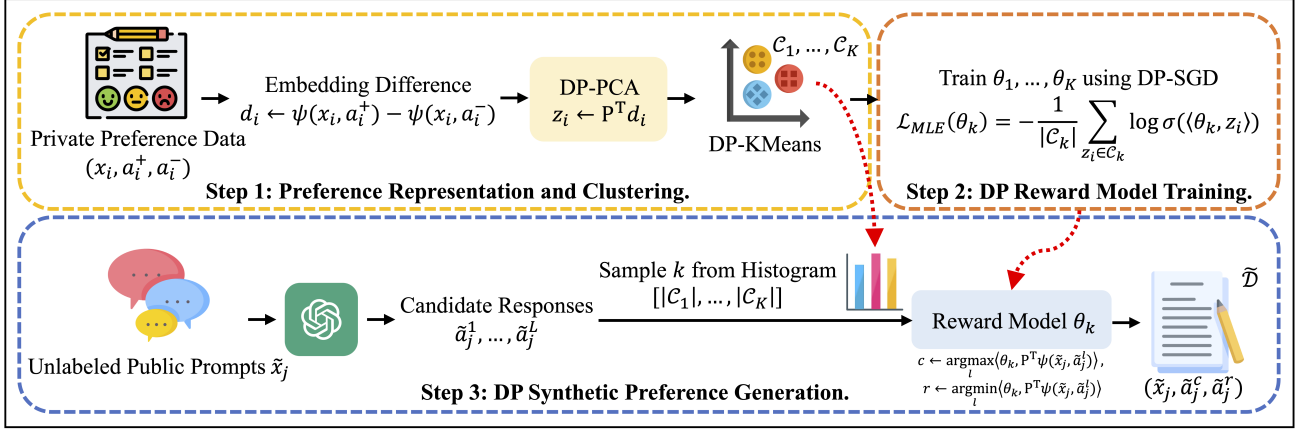


Figure 1. Overview of DPPrefSyn. The framework synthesizes DP preference data through three coordinated stages: 1) representing preference samples as embedding differences and clustering them with DP-PCA and DP-KMeans, 2) training cluster-specific DP reward models via DP-SGD, and 3) generating synthetic preference samples from public prompts under guidance from the DP-protected distribution of reward functions.

generate synthetic text for supervised instruction tuning.

Because DP fine-tuning can be computationally expensive and, in some cases, infeasible for non-public language models, a more recent paradigm studies DP synthetic data generation through LLM API prompting (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These approaches typically generate candidate synthetic samples through API access and then, under privacy-preserving selection rules, retain outputs that resemble the private data. For instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples according to embedding-space similarity with private examples. Our work follows this API-oriented synthetic-data paradigm while specializing it to LLM preference alignment.

Diversity in Human Preferences. A growing body of work emphasizes that human preferences are intrinsically heterogeneous (Denton et al., 2021; Aroyo et al., 2023a,b; Chakraborty et al., 2024). Salient sources of variation include sociodemographic differences (e.g., race, gender, age), personal beliefs and biases, heterogeneous levels of domain expertise, and the inherent ambiguity of natural language (Sandri et al., 2023; Vogels, 2021). Motivated by this observation, recent studies have explored personalized preference alignment as a means of better reflecting the values of distinct user groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For example, Singh et al. (2025) develop a meta-learning framework in which an LLM adapts to individual users using a small number of preference examples. In contrast, our objective is to generate DP synthetic preference data that captures the underlying diversity of human feedback; as shown in Section 5.3, explicitly modeling this diversity improves alignment performance relative to approaches that collapse preferences into a homogeneous signal.

3. Problem Definition

Preference Alignment of LLMs. We study LLM preference alignment from pairwise human feedback, represented by a preference dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each observation contains a prompt x_i and two candidate responses a_i^+, a_i^- , with human annotators indicating that a_i^+ is preferred to a_i^- .

Following prior work on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we formalize human preferences through the Bradley-Terry model (Bradley & Terry, 1952):

$$\begin{aligned} \mathbb{P}[a_i^+ \succ a_i^- \mid x_i] &= \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} \\ &= \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)), \end{aligned}$$

where $a_i^+ \succ a_i^-$ denotes the event that a_i^+ is preferred to a_i^- , $r^*(x, a)$ is a latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

This probabilistic formulation underlies both RLHF, which explicitly learns a reward model from preference data, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. Our goal is to protect the private preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ from an adversary who attempts to access or infer sensitive information about individual prompts, responses, or preference labels from the outputs of a fine-tuned LLM. To this end, we generate a DP synthetic preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that preserves the preference-relevant structure of $\mathcal{D}_{\text{priv}}$ while obfuscating individual records. By the post-processing property of DP, $\tilde{\mathcal{D}}$

can be reused across different preference-alignment algorithms and LLM backbones without incurring any additional privacy cost.

We use the following standard definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Human preferences in private datasets are typically not homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Different annotators may emphasize distinct response attributes, including factual accuracy, politeness, creativity, or clarity, thereby producing diverse and occasionally conflicting signals in the observed data. Consequently, faithful synthesis of $\tilde{\mathcal{D}}$ requires preservation of the heterogeneous preference structure embedded in $\mathcal{D}_{\text{priv}}$.

To make this preservation tractable, we posit a linear structure for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt the linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ denotes the *shared* feature representation and $\theta \in \mathbb{R}^d$ denotes a *user-dependent* parameter.

This linear structure provides a useful compromise between *expressiveness* and *tractability* when modeling diverse user preferences. Although individual users may express heterogeneous preferences, their reward functions can share common semantic features in a learned embedding space (i.e., $\phi(x, a)$), differing primarily in the weights assigned to those features (i.e., θ). The resulting linear parameterization therefore captures shared semantic structure while permitting variation across users or preference clusters, which is especially valuable when data are limited and privacy constraints are strong. As the experiments in Section B.6 demonstrate, this linear reward structure can effectively capture preference signals and achieves competitive performance relative to a fully fine-tuned GPT-2 reward model despite its simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before presenting the full procedure, we identify several algorithmic components that address the principal challenges in differentially private preference-data synthesis.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model implies that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

Consequently, for a tuple (x_i, a_i^+, a_i^-) , the event $a_i^+ \succ a_i^-$ implies $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$, meaning that the preference vector θ is strongly aligned with $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. For human annotators who express similar preferences, the feature-difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$ across their rated pairs are therefore expected to occupy similar directions in the embedding space. This intuition motivates clustering user preference samples according to the feature-difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$, so that preferences within a cluster can be well approximated by a shared reward function parameterized by a common preference vector θ . In Section B.9, we demonstrate that the clusters obtained in this manner are meaningful and interpretable in terms of user preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural instantiation of the feature vector $\phi(x, a)$ is the embedding of the corresponding (prompt, response) pair, since embeddings encode semantic similarities and differences that directly inform human preference judgments. These embeddings, however, are typically high dimensional (e.g., 1024 dimensions) because they are designed to capture rich semantic and stylistic variation. Learning in such a high-dimensional space generally requires many samples, whereas each preference cluster contains only a limited subset of the private data. To mitigate this sample-efficiency bottleneck, we apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to reduce embedding dimensionality and obtain compact feature representations for clustering and reward-model training. This step improves sample efficiency and clustering stability while preserving the essential preference signals required for effective alignment.

Utilizing Public Prompts to Save DP Budget. Simultaneously synthesizing prompts, preferred responses, and less preferred responses can rapidly exhaust the privacy budget and degrade downstream utility. We therefore use public prompts, which avoids spending privacy budget on DP prompt synthesis and permits the full budget to be directed toward modeling preferences over responses. For each public prompt, we generate multiple candidate responses with an LLM and apply a DP-trained distribution of reward functions, learned from private preference data, to construct preference pairs. Although public prompts may differ distributionally from private prompts, our experiments show that the resulting synthetic preference data preserves the statistical preference distribution of the private dataset and remains effective for downstream alignment despite this prompt shift.

4.2. DPPrefSyn Algorithm

We now introduce DPPrefSyn (Algorithm 1), which constructs a DP synthetic preference dataset from the private dataset $\mathcal{D}_{\text{priv}}$. The algorithm consists of three main stages:

Step 1: Preference Representation and Clustering.

We encode each prompt-response pair in $\mathcal{D}_{\text{priv}}$ using a public sentence embedding model ψ . For each sample (x_i, a_i^+, a_i^-) , we concatenate the prompt with each response to form $[x_i; a_i^+]$ and $[x_i; a_i^-]$, compute their embeddings through ψ , and define the resulting preference-direction vector as $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

To reduce dimensionality privately, we apply DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, obtaining a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ under ε_0 -DP (Line 5). At a high level, DP-PCA (Amin et al., 2019) approximates the eigendecomposition of the data covariance matrix by estimating eigenvalues and eigenvectors separately under DP. Each d_i is then projected into a lower-dimensional representation $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). We apply DP-KMeans clustering (Su et al., 2016) to $\{z_i\}_{i=1}^n$ so that samples with similar preference patterns are grouped under ε_1 -DP, yielding K clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ in Line 7. In essence, DP-KMeans (Su et al., 2016) is a differentially private variant of Lloyd’s algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, we estimate a shared preference vector $\theta_k \in \mathbb{R}^p$ that captures the cluster-specific preference pattern by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i is the DP-PCA-projected embedding difference from Step 1. Each reward model is trained with DP-SGD, a DP variant of SGD (Abadi et al., 2016), using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. We compute a histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ recording the number of samples in each cluster (Line 11) and normalize it into a cluster distribution $\mathbf{p}$, namely $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12).

For each public prompt $\tilde{x}_j$, we prompt an LLM at a relatively high temperature to generate L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). We then sample a cluster index $k \sim \mathbf{p}$ according to the DP histogram (Line 15) and evaluate the candidates $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ with the reward model parameterized by θ_k . Specifically, we compute embeddings $\psi(\tilde{x}_j, \tilde{a}_j^l)$ for $l = 1, \dots, L$, project them into the DP-PCA subspace using $\mathbf{P}$, and assign each response $\tilde{a}_j^l$ the reward value $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. We then select the responses with the highest and lowest reward values as the preferred and less preferred responses, respectively: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). If the re-

ward gap is too small (e.g., < 0.5), we discard the sample to maintain preference quality; otherwise, we add $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ to $\tilde{\mathcal{D}}$. Repeating this procedure for all public prompts produces the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
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Privacy Analysis of DPPrefSyn. We allocate privacy budget ε_0 to DP-PCA (Line 5) and ε_1 to DP-KMeans (Line 7), while using the remaining budget to train per-cluster reward models with DP-SGD and noise multiplier σ_2 (Line 9). For DP reward-model training, we invoke parallel composition: because the clusters are disjoint, modifying one training example affects only one cluster, so the total privacy cost is bounded by the cost of training on the smallest cluster. We guarantee a known lower bound on cluster size by discarding clusters with too few samples. We then use the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the privacy costs of the DP-SGD steps, ensuring that the reward-model training component satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. All remaining operations involve only public data or DP post-processing and therefore incur no additional privacy cost. We provide the complete proof and privacy-accounting code in Section A.

Discussion on Prompt Distribution Shift. DPPrefSyn remains effective even when the prompt distributions of the private and public datasets differ. The essential point is that diverse user preferences are encoded through θ_k , the parameter of the linear reward model, rather than through $\phi(x, a)$, the feature vector tied to a specific prompt-response pair.

Intuitively, a human user may interact with an LLM through many different prompts while maintaining relatively stable underlying preferences. Thus, as long as the distribution of θ_k resembles that in the private dataset, the resulting synthetic preferences can continue to reflect private preference patterns. DPPrefSyn enforces this correspondence by randomly sampling a cluster index k according to the DP histogram (Line 15) for each public prompt and then selecting synthetic response pairs accordingly. Consequently, even when private and public prompt distributions differ, DPPrefSyn can preserve the preference patterns captured in the private data.

5. Experiments

Datasets. We evaluate DPPrefSyn on both question-answering and summarization tasks. For question answering, we use OpenAssistant (Köpf et al., 2023), which contains assistant-style conversations, and Anthropic-HH (Bai et al., 2022), which provides human preference comparisons centered on helpfulness and harmlessness. For summarization, we use TL;DR (Stiennon et al., 2020), which contains human preference annotations over pairs of summaries. To instantiate public data, we use prompts from Alpaca (Taori et al., 2023) for OpenAssistant QA, SafeRLHF (Ji et al., 2024) for Anthropic-HH QA, and XSum (Narayan et al., 2018) for TL;DR summarization. Additional dataset details appear in Section B.2.

Setup for Algorithm 1. Unless otherwise specified, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We set the DP-PCA projected dimension to $p = 20$ and the number of DP-KMeans clusters to $K = 5$. For DP-SGD, we use a learning rate of 0.1, a batch size of 4, 4 training epochs, and a gradient clipping norm of 1.0. Candidate responses are generated by the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) at temperature 0.9, with $L = 5$ candidates per prompt. We evaluate overall privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$ and set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We examine the effects of alternative generator models in Table 3, and we study embedding models and hyperparameters in Section 5.3.

Evaluation. For downstream preference alignment, we fine-tune Pythia-2.8B (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), using the preferred response in each preference pair as the training target, and then apply DPO (Rafailov et al., 2024) to further optimize the SFT model on preference pairs. Following Rafailov et al. (2024), we evaluate the win rate of model-generated responses against the preferred test-set responses using GPT-4o. Additional downstream models and preference-alignment procedures are examined as ablations in Section 5.2.

Baselines. We compare DPPrefSyn with DP-SGD-based

fine-tuning of the downstream model on real data (Abadi et al., 2016), denoted DP-FT. DPPrefSyn is more reusable and versatile than DP-FT because it produces a DP synthetic dataset that can train different models under different alignment algorithms without additional privacy cost, whereas DP-FT expends the privacy budget on the training of a single privatized model. We also include a fully private baseline with $\varepsilon = 0$, corresponding to evaluation of the base model without preference fine-tuning.

Further details on experimental setups, hyperparameters, metrics, and baselines are provided in Section B.3.

5.1. Understanding the Performance of DPPrefSyn

We first analyze the performance of DPPrefSyn through two research questions concerning its privacy-utility trade-off. The principal results are reported in Table 1.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? Across all tasks and privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$, DPPrefSyn consistently yields a stronger privacy-utility trade-off than DP-FT, and, while preserving privacy, it surpasses the utility of fine-tuning on real private data without DP constraints (DP-FT with $\varepsilon = \infty$). On OpenAssistant QA with $\varepsilon = 2$, for example, DPPrefSyn obtains 9.10% after SFT and 11.04% after DPO, whereas DP-FT reaches only 4.38% after SFT and 6.18% after DPO. Notably, DPPrefSyn also exceeds the non-private real-data baseline (DP-FT with $\varepsilon = \infty$), which achieves 4.75% after SFT and 8.20% after DPO. The same pattern appears on Anthropic-HH with $\varepsilon = 2$: DPPrefSyn achieves 55.21% after SFT and 56.48% after DPO, compared with DP-FT’s 31.53% and 37.02%, respectively, and it also outperforms direct non-private fine-tuning on real data, which obtains 31.98% after SFT and 38.72% after DPO.

These results reveal two **key advantages** of the proposed DP synthetic data. **First**, even SFT alone benefits substantially from DPPrefSyn relative to direct SFT on private data. On TL;DR with $\varepsilon = 2$, for instance, DPPrefSyn reaches a 62.49% SFT win rate, far above the 22.57% obtained by SFT on private data. This improvement arises primarily because DPPrefSyn uses a high-performing LLM to generate candidate responses, thereby producing high-quality supervision for SFT. The advantage persists even when public data are scarce: on OpenAssistant QA with $\varepsilon = 4$ and only 5K public prompts, DPPrefSyn attains 6.80% after SFT, compared with 4.46% for direct SFT on the approximately 14K-sample private dataset (Table 4), demonstrating its capacity to alleviate data scarcity while preserving utility. **Second**, applying DPO after SFT with our synthetic data yields additional gains, indicating that the synthetic preference pairs encode meaningful distinctions aligned with human judgments. These distinctions are induced by private

Table 1. GPT-4o win rates on test sets for the downstream Pythia-2.8B model after SFT and DPO. $\varepsilon = 0$ denotes the base, fully private, non-finetuned LLM. *Relative to DP-FT, DPPrefSyn attains higher win rates at privacy levels $\varepsilon = 0.5, 1, 2, 4, 8$ and even exceeds the utility of non-private fine-tuning (DP-FT with $\varepsilon = \infty$).* Results report mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. On OpenAssistant, DPPrefSyn attains higher win rates than private-data fine-tuning across SFT, DPO, and PPO while maintaining strong privacy guarantees. Results report mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

reward functions that capture diverse user preferences under DP. Although the incremental DPO gain is smaller than in direct private-data fine-tuning, likely because the SFT model trained on synthetic data is already strong, the aggregate fine-tuning improvement obtained from the DP synthetic dataset remains substantially larger than that obtained from real private data.

RQ2: How does DPPrefSyn perform across different privacy budget ε ? **First**, Table 1 shows that DPPrefSyn generally improves as ε increases from 0.5, 1, 2, 4, and 8 to ∞ , while maintaining a performance advantage over DP-FT, suggesting favorable scalability with respect to the privacy budget. **Second**, in some settings DPPrefSyn is more robust to DP noise than DP-FT. For example, on OpenAssistant with Llama-3-8B as the downstream model, DP-FT’s DPO performance decreases from 59.13% to 45.79% when ε is reduced from ∞ to 4, whereas DPPrefSyn decreases only from 61.52% to 60.25% (see Table 3 in Section 5.2). A plausible explanation is that DP-FT directly perturbs model parameters through DP-SGD, while DPPrefSyn injects privacy noise during data synthesis in a way that better preserves downstream utility.

5.2. Understanding the Properties of DPPrefSyn

We next examine several properties of DPPrefSyn, including its transferability across preference-alignment methods and LLM backbones, its behavior under different quantities of public data, and its empirical robustness to privacy attacks.

Table 3. Across downstream LLMs on OpenAssistant, DPPrefSyn yields higher post-DPO win rates than private-data fine-tuning while preserving strong privacy guarantees; it also benefits from stronger generator LLMs.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

RQ3: Is DPPrefSyn effective across different preference alignment methods? **DPPrefSyn is compatible with multiple preference-alignment methods and remains effective under strong DP constraints.** In Table 2, we evaluate DPPrefSyn with SFT, DPO, and PPO. For PPO, we train K reward models, one per cluster, to represent diverse preference patterns; during PPO training, we average the outputs of these K reward models and use the resulting scalar as the policy-optimization reward. Across SFT, DPO, and PPO, DPPrefSyn consistently provides higher utility than fine-tuning on private data while preserving strong privacy guarantees.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? **DPPrefSyn remains effective across a broad set of downstream LLMs.** In Table 3, we evaluate downstream models from several families, including Qwen-3-4B, Gemma-2-2B, and Llama-3-8B, using Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generators. After DPO, DPPrefSyn outperforms real-data fine-tuning across these model families while maintaining privacy. For instance, when Qwen3-4B-Instruct-2507 is used as the generator, DPPrefSyn achieves an 86.52% win rate on Qwen-3-4B at $\varepsilon = 4$, substantially

Table 4. DPPrefSyn achieves higher win rates than private-data fine-tuning across varying public-data sizes m . Results report mean and standard deviation over 5 random seeds.

	DPPrefSyn ($\epsilon = 4$)			DP-FT ($\epsilon = 4$)	DP-FT ($\epsilon = \infty$)
Data Type (Size)	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

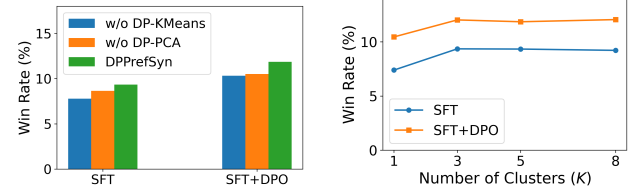
Table 5. On OpenAssistant, DPPrefSyn yields lower AUC scores against MIAs (Feng et al., 2025), approaching random guessing. Results report mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64 _{1.29}	49.98 _{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51 _{1.38}	50.43 _{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45 _{0.87}	50.67 _{0.96}

exceeding non-private real-data fine-tuning at 59.41%. We also observe that stronger generator models improve DPPrefSyn, with Qwen3-4B-Instruct-2507 outperforming Llama-3-8B-Instruct.

RQ5: Can DPPrefSyn be effective with varying scales of public data? We study the effect of public-data size m in Table 4. The private OpenAssistant dataset contains 14K preference annotations, whereas the full public Alpaca dataset contains 52K prompts; from this public pool, we randomly sample subsets of sizes $m = 5K, 10K, 52K$ and use DPPrefSyn to generate synthetic preference data. Even with only 5K public prompts, DPPrefSyn reaches 6.80% after SFT and 9.44% after DPO under $\epsilon = 4$, outperforming the DP-FT baseline values of 4.46% and 6.12%, respectively. These results show that **DPPrefSyn remains effective even when public data are limited**, owing to high-quality LLM-generated responses and private reward functions that accurately capture user preferences. Moreover, **as the public-data size increases, the performance margin over the baseline becomes more pronounced**, indicating that DP-PrefSyn can exploit abundant public prompts for preference alignment and thereby further mitigate scarcity in private preference data.

RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? In Table 5, we conduct membership inference attacks (MIAs) (Shokri et al., 2017) against DPO-fine-tuned downstream models on OpenAssistant and report AUC scores. Following Feng et al. (2025), who instantiate an MIA for preference data in LLM alignment, the attack objective is to determine whether a private sample (x, a^+, a^-) was used during preference alignment. We find that DP-FT does not sufficiently mitigate MIAs, whereas **DPPrefSyn**



(a) Ablation removing DP-KMeans or DP-PCA.

(b) Ablation over the number of clusters.

consistently exhibits lower AUC scores, close to random guessing. This behavior suggests that DPPrefSyn is more robust to empirical privacy attacks, plausibly because fine-tuning on synthetic data reduces the tendency to overfit individual private records. Additional details are deferred to Section B.4.

5.3. Validating the Design of DPPrefSyn

We further validate DPPrefSyn by isolating and analyzing its principal algorithmic components.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? We assess this question on OpenAssistant at $\epsilon = 4$ by removing either DP-KMeans or DP-PCA and by varying the number of clusters K . **(i) Removing either DP-KMeans or DP-PCA.** As shown in Figure 2a, eliminating either component degrades performance. Without DP-KMeans, all samples collapse into a single cluster, while removing DP-PCA produces ineffective clustering and sample-inefficient learning in the original high-dimensional space. Using multiple clusters together with DP-PCA therefore improves performance relative to the single-cluster alternative. **(ii) Varying the number of clusters K .** In Figure 2b, we evaluate $K = 1, 3, 5, 8$. Performance initially improves at $K = 3$ and then remains comparatively stable at $K = 5, 8$, suggesting that DPPrefSyn is not highly sensitive to K and that moderate values, such as $K = 3$ or $K = 5$, provide a favorable balance between preference diversity and data efficiency. Overall, these results support the benefit of representing diverse preferences through multiple clusters.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? In Section B.5, we ablate the embedding model ψ , the DP-PCA dimension p , the number of candidates per prompt L , the filtering threshold, and the privacy-budget allocation. Across sentence encoders, including bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DPPrefSyn remains consistently effective. We find that $p = 20$ yields the best overall post-DPO performance, $L = 5$ is a strong default, and a moderate filtering threshold of 0.5 slightly outperforms no filtering, whereas a threshold of 1 is overly aggressive and harms performance. DPPrefSyn

is also robust to reasonable changes in the privacy-budget allocation between ϵ_0 and ϵ_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To rule out these potential confounders, we conduct controlled ablations in which all methods share either the same prompt set or the same response generator. Under these matched conditions, DPPrefSyn continues to outperform DP-FT (Appendix B.7), indicating that **its performance gains do not merely arise from higher-quality public prompts or stronger response generators.**

6. Conclusion

This work introduced DPPrefSyn, an algorithm for synthesizing DP preference data from a private preference corpus. By modeling preference diversity through clustering, improving statistical efficiency with DP-PCA, and conserving privacy budget through the use of public prompts, DPPrefSyn enables effective privacy-preserving preference-data synthesis for LLM alignment. The experiments demonstrate that DPPrefSyn provides formal privacy guarantees while outperforming fine-tuning on real private data.

7. Limitations

DPPrefSyn requires LLM API access for preference-data synthesis, which may introduce additional API costs. The present study focuses on English text-based preference data, leaving extensions to multilingual settings and multimodal data as important directions for future work. In addition, because of computational constraints, our experiments cover models only up to 8B parameters.

Impact statement

This work supports preference alignment for large language models without direct exposure of personal information, thereby helping reconcile the benefits of large-scale data with the imperative of individual privacy protection, promoting more ethical data use, and strengthening societal trust in increasingly data-driven systems.

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A. Privacy Analysis

This section provides the privacy analysis for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We first introduce the concepts and supporting results from the literature that are required for the analysis.

Many DP mechanisms, including the Gaussian mechanism, provide a family of (ε, δ) -DP guarantees. More precisely, for each fixed ε , there exists a corresponding $\delta(\varepsilon)$ such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Privacy curves are advantageous because they permit tighter composition guarantees than classical advanced composition theorems (Dwork et al., 2014). In particular, privacy curves enable numerical composition, which yields the tightest guarantees. Gopi et al. (2021) provide privacy curves for the composition of several standard mechanisms, including the Gaussian mechanism and the subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{\text{error}}}{k}\right)\right\}.$$

Gopi et al. (2021) also characterize the privacy loss of a subsampled mechanism in terms of the privacy loss of the original mechanism, a result that can be used to bound the privacy loss of DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We now proceed to the privacy analysis of our algorithm.

Proof. The algorithm contains three components that access the private dataset: the DP-PCA subroutine satisfying ε_0 -DP (Amin et al., 2019) in Line 5, the DP-KMeans subroutine satisfying ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. In the experiments, we allocate ε_0 to DP-PCA and ε_1 to DP-KMeans, with the remaining privacy budget assigned to DP-SGD and composed using the Privacy Random Variable accountant (Gopi et al., 2021).

In Line 7, the low-dimensional embeddings $\{z_i\}_{i=1}^n$ are partitioned under DP into K disjoint groups $\mathcal{C}_1, \dots, \mathcal{C}_K$. A separate linear model is then trained on each cluster using DP-SGD in Line 9. By the parallel composition property of DP, the overall privacy cost of this stage depends only on the smallest cluster used for training. To ensure a known lower bound on the

sample size for each model, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K+4)$ samples. For DP-SGD, we use batch size 4, 4 training epochs, and gradient clipping norm 1.0. The per-iteration privacy loss follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4); we then invoke Theorem A.3 to compose the DP-SGD iterations (Gopi et al., 2021), ensuring that this component satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. The DP parameters ε_0 , ε_1 , and σ_2 used in our experiments are reported in Table 7.

The privacy-accounting code is provided below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):
    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )
    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )
    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )
    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are conducted on NVIDIA H100 GPUs, each equipped with 80 GB of memory. DPPrefSyn is run on a single NVIDIA H100 GPU, and completion requires several hours depending on dataset size.

B.2. Datasets

This section describes the datasets used in the experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated and human-annotated assistant-style conversation corpus containing 161,443 messages across 35 languages, 461,292 quality ratings, and more than 10,000 fully annotated conversation trees. In our experiments, we use only the English subset, which contains 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference annotations used for RLHF reward-model training. For helpfulness, the data are divided into train and test sets across three tranches: outputs from base models (context-distilled 52B), samples filtered through rejection sampling with a preference model, and data collected during online training. For harmlessness, the data are collected only from base models but retain the same format. The dataset contains 161K training and 8.55K test comparisons; for evaluation, we sample 1,000 examples from the test set.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) contains human preference annotations over pairs of summaries, with annotators asked to select the better of two summaries. It was originally used to train a reward model for aligning summarization behavior with human preferences. The summaries are drawn from TL;DR, and additional validation data are drawn from CNN and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 examples from the validation split as the test set.

- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated through self-instruct with OpenAI’s text-davinci-003. Each example contains an instruction and a corresponding response, with the dataset designed to improve language models’ instruction-following behavior. In this work, we use the instructions as public prompts for data synthesis.
- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset containing both performance and safety preferences. It includes constraints across more than ten dimensions, including insults, immorality, crime, emotional harm, and privacy, and is designed for fine-grained value alignment in RLHF. In this work, we use the 73.9K prompts from the training split as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) is designed for evaluating abstractive single-document summarization systems, where the task is to generate a one-sentence summary answering the question, “What is the article about?” The dataset contains 226,711 BBC news articles from 2010 to 2017, each paired with a concise summary, and spans domains such as politics, sports, business, science, and health. It includes 204,045 training, 11,332 validation, and 11,334 test examples. In this work, we use documents from the training split as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

This section specifies the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. In DPPrefSyn, for each preference sample (x_i, a_i^+, a_i^-) , we concatenate the prompt with each response to form two texts, $[x_i; a_i^+]$ and $[x_i; a_i^-]$. The prompt-response concatenation formats used in the experiments are shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . The DP-PCA implementation is taken from Shoemate et al. (2021) and is based on the algorithm of Amin et al. (2019); the DP-KMeans implementation is taken from Holohan et al. (2019) and follows Su et al. (2016). We set the DP-PCA projected dimension to $p = 20$ and the number of DP-KMeans clusters to $K = 5$. During reward-model training with DP-SGD, we discard clusters containing fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples so that each model has a known sample-size lower bound. For DP-SGD, we use the SGD optimizer with learning rate 0.1, batch size 4, 4 epochs, and gradient clipping norm 1.0, implemented with Opacus (Yousefpour et al., 2021).

By default, candidate responses for each public prompt are generated with the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023). We set the number of candidates per prompt to $L = 5$ and use generation temperature 0.9. The prompt formats for multiple-completion generation are shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

```
Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:
```

Privacy Parameters in DPPrefSyn. In all experiments, we set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. The values of $|\mathcal{D}_{\text{priv}}|$ are reported in Table 6, and the DP parameters ε_0 , ε_1 , and σ_2 are reported in Table 7.

Table 6. Sizes of the private datasets used in the experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 , and σ_2 used in the DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first apply supervised fine-tuning (SFT), using the preferred response in each preference pair as the training target, and then apply DPO (Rafailov et al., 2024) to further fine-tune the SFT model on preference pairs. All models are trained with bfloat16 precision. Following Rafailov et al. (2024), we use learning rate 5×10^{-7} and train for 1 epoch for SFT; for DPO, we use $\beta = 0.1$, learning rate 1×10^{-6} , and train for 3 epochs. The batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

We use win rate as the utility metric, measuring how often the fine-tuned model’s response is preferred over a reference response. For each test prompt, we generate a response from the fine-tuned model at temperature 0.0 and compare it with the preferred response in the test set using GPT-4o as the evaluator. The win rate is the percentage of comparisons in which GPT-4o selects the model response as better.

We adopt win rate because it follows the standard evaluation protocol in recent preference-alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM serves as an automated judge for estimating human preference judgments. Prior work indicates that high-quality frontier models provide reliable and reproducible judgments that correlate strongly with human annotations, while offering greater scalability and consistency than human studies (Zheng et al., 2023). We use GPT-4o as the judge because it was the most advanced model available at the time of our experiments. The prompt formats used to generate win rates are provided below, following Rafailov et al. (2024).

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format: Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, we implement DP-SGD with Opacus (Yousefpour et al., 2021), FSDP2, and Ghost Clipping. To simplify privacy accounting, we assume that each user contributes to either SFT or DPO by evenly splitting the private dataset into two disjoint halves. Privacy loss is tracked with the PRV accountant. We conduct a grid search over clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and use 0.1 in the final experiments. Learning rates are searched over $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$; the selected values are 5×10^{-5} for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, while 5×10^{-6} is used for Llama-3-8B. The batch size is 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 SFT epochs and 3 DPO epochs.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

Although DP provides a formal guarantee against privacy leakage, empirical privacy risks remain important to assess. We therefore perform membership inference attacks (MIAs) (Shokri et al., 2017) against DPO-fine-tuned downstream models on OpenAssistant and report AUC scores in Table 5.

We follow Feng et al. (2025), who instantiate an MIA for preference data in LLM alignment. The attack objective is to determine whether a private sample (x, a^+, a^-) was used during preference alignment. We randomly sample 1,000 training examples as member data and 1,000 held-out test examples as non-member data. DP-FT does not sufficiently mitigate MIAs, whereas DPPrefSyn consistently produces lower AUC scores close to random guessing. This suggests that DPPrefSyn is more robust to empirical privacy attacks, plausibly because synthetic-data fine-tuning reduces overfitting to individual private records.

B.5. Ablation Studies on Hyperparameters

This section presents ablation studies on the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filtering threshold, and privacy-budget allocation. All experiments are conducted on OpenAssistant with $\epsilon = 4$.

Varying the embedding model ψ . We evaluate several sentence encoders, including BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). The results in Table 8 indicate that, although embedding quality affects performance, DPPrefSyn remains effective across different embedding models.

Table 8. Performance of DPPrefSyn under different embedding models ψ on OpenAssistant at $\epsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . We evaluate $p = 10, 20, 30$, as reported in Table 9. The results show that $p = 20$ provides the best overall performance after DPO, making it a suitable default choice.

Table 9. Performance of DPPrefSyn under varying DP-PCA dimensions p on OpenAssistant at $\epsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . We evaluate $L = 3, 5, 8$, as shown in Table 10. Performance improves when L increases from 3 to 5, whereas $L = 8$ yields no further gains, indicating that $L = 5$ is an effective default choice.

Table 10. Performance of DPPrefSyn under varying numbers of candidates per prompt L on OpenAssistant at $\epsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. We evaluate thresholds of 0, 0.5, and 1, as shown in Table 11. A moderate threshold of 0.5 slightly outperforms no filtering because it removes noisy or uninformative preference pairs, whereas a threshold of 1 is overly aggressive, reduces data coverage, and causes a performance decline.

Table 11. Performance of DPPrefSyn under varying filtering thresholds on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We evaluate alternative budget splits between DP-PCA (ε_0) and DP-KMeans (ε_1). In the default setting, under total privacy budget $\varepsilon = 4$, we allocate $\varepsilon_0 = \varepsilon_1 = 0.5$. We additionally test two allocations on OpenAssistant: (i) assigning more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$), and (ii) assigning more budget to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). The results in Table 12 show that DPPrefSyn is robust to reasonable variation in the budget split.

Table 12. Performance of DPPrefSyn under varying privacy-budget allocations on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

This section discusses the design choice of using linear reward models in DPPrefSyn (Line 9). The choice follows the Bradley–Terry model with a linear reward function, a widely adopted assumption in the preference-alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023), and is formalized in Section 3.

To assess the validity of this modeling choice, we empirically compare our linear model with a fully fine-tuned GPT-2 reward model trained on Anthropic-HH. For each test triplet (prompt, preferred response, dispreferred response), both responses are concatenated with the prompt and passed into the trained reward model. The model assigns a scalar value to each input, and the prediction is counted as correct when the preferred response receives the higher value. Accuracy is computed as the number of correct predictions divided by the total number of triplets. The linear model achieves 66% accuracy, whereas the fully fine-tuned GPT-2 model achieves 62%, showing that the linear reward model captures preference signals effectively and remains competitive despite its simplicity.

B.7. Ablation Studies on Prompt and Response Generator Quality

This section evaluates how prompt quality and response-generator quality affect DPPrefSyn. All experiments are conducted on OpenAssistant with $\varepsilon = 4$.

Prompt Quality. We test whether DPPrefSyn’s gains are attributable to higher-quality public prompts. Specifically, we rerun DPPrefSyn using OpenAssistant prompts as the public prompt set instead of Alpaca. The DP-FT baseline continues to use its original private OpenAssistant dataset, so both methods rely on *the same set of prompts* from OpenAssistant. The results are reported in Table 13.

Table 13. Prompt-quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

The results show that DPPrefSyn still outperforms DP-FT when both methods use the same OpenAssistant prompts, indicating that its gains are not solely attributable to higher-quality public prompts.

Candidate Response Quality. To test whether DPPrefSyn’s gains arise from stronger response generators, we include an enhanced DP-FT baseline. We first use *the same LLaMA-7B-Chat generator* as DPPrefSyn to produce candidate responses for each prompt, then rank those responses with a trained reward model to construct synthetic preference pairs. Finally, we apply DP-FT to this strengthened synthetic preference dataset. This design ensures that the enhanced DP-FT baseline receives responses of comparable quality to those used by DPPrefSyn. Results are provided in Table 14.

Table 14. Response-quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Although the enhanced DP-FT baseline improves over the original DP-FT, DPPrefSyn still achieves higher performance, indicating that its gains cannot be explained solely by the use of a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We conduct a direct head-to-head comparison on OpenAssistant in which GPT-4o judges DPPrefSyn outputs against DP-FT outputs. The win rate of DPPrefSyn over DP-FT is reported in Table 15.

DPPrefSyn is preferred to DP-FT in more than 70% of cases across privacy budgets, corroborating the consistent advantage observed in the main results.

Table 15. DPPrefSyn win rate over DP-FT on OpenAssistant.

ε	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

This section assesses whether the clusters produced by DPPrefSyn correspond to meaningful preference groups. Because the datasets used in our experiments do not include information about specific human annotator groups, systematically interpreting the preference style represented by each DP-KMeans cluster is difficult. To address this limitation, we simulate two groups of human annotators with different preferences using GPT-4, following Chakraborty et al. (2024). Specifically, for each prompt from Alpaca (Taori et al., 2023), GPT-4 generates two responses under different instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with the two responses, together with a simulated preference label and annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans with $\varepsilon = \infty$ to cluster the embedding differences of the paired responses. The resulting clusters align strongly with the simulated preference groups: one cluster is dominated by PhD student preferences (95.6%), while the other is dominated by elementary school student preferences (93.1%). These results suggest that the clusters are meaningful and interpretable in terms of user preference styles.

B.10. Samples from DPPrefSyn

In the section, we present generated samples using DPPrefSyn. Representative examples for each task are provided in Table 16. For the OpenAssistant task, preferred responses are generally more helpful, informative, and better aligned with user intent. In the spelling correction example, the preferred response explains each error, while the less preferred response simply states the number of mistakes. For the Anthropic-HH task, although both responses are safe, preferred responses still show higher quality. In the adoption prompt, the preferred response addresses both legal and emotional aspects, encourages seeking professional help, and emphasizes child support, while the less preferred one is shorter and

Table 16. Generated samples from DPPrefSyn with $\varepsilon = 4$. In the OpenAssistant example, the preferred response is more helpful and better aligned with user intent; in the Anthropic-HH example, the preferred response is more supportive and informative; and in the TL;DR example, the preferred summary is more concise and more focused on the central point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

more formal, lacking empathy and practical guidance. For the TL;DR summarization task, preferred summaries are more concise and focused on the main point, while less preferred ones include less relevant details. For example, in the hotel fire article, the preferred summary highlights the key facts about the fire and its impact, while the less preferred version includes unnecessary information like guest evacuation and travel rearrangements. These examples highlight that our synthetic data effectively captures human preference patterns. We provide more samples from DPPrefSyn below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\varepsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4. Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* "Finnished" should be "finished"

* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
 2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night's sleep.
 3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
 4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
 5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
 6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
 7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
 8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
 9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
 10. Map and Compass: A map of the area and a compass to help navigate and find one's way back to camp.
- I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent
 2. Sleeping bag
 3. Sleeping pad
 4. Backpack
 5. Flashlight or headlamp
 6. Water bottle or hydration bladder
 7. Portable stove or camping grill
 8. Cooking pot and utensils
 9. Toiletries (toothbrush, toothpaste, wipes, etc.)
 10. First aid kit
- Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

- * Oak tree: Plant
- * Copper ore: Mineral
- * Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I'd be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\varepsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can

have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\varepsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been "no hostage crises", adding that the man was "alone in the room".

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun. "His firearm was taken away," Dr Kutlubay said, adding that the gun in the officer's possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year's Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer's possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the "investigation is now being conducted by the IPCC".

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It's the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.